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import streamlit as st
import pandas as pd
import plotly.express as px

# Set up the page layout
st.set_page_config(page_title="Revenue Dashboard", layout="wide")
st.title("Revenue Analysis Dashboard")

# 1. LOAD DATA (Currently using mock data so you can test it immediately)
# To use your own data, delete the mock data and uncomment the line below:
# df = pd.read_csv("your_data.csv") OR pd.read_excel("your_data.xlsx")

df = pd.read_excel("data.xlsx")
df['Date'] = pd.to_datetime(df['Date'])

# 2. FILTERS & SLICERS (Sidebar)
st.sidebar.header("Dashboard Filters")

# Create a multi-select dropdown for products
product_list = df['Product'].unique()
selected_products = st.sidebar.multiselect(
    "Filter by Product",
    options=product_list,
    default=product_list
)

# Apply the filter to the dataframe
filtered_df = df[df['Product'].isin(selected_products)]

# DATE FILTER (ADD HERE)
date_range = st.sidebar.date_input(
    "Select Date Range",
    [df['Date'].min(), df['Date'].max()]
)

filtered_df = filtered_df[
    (filtered_df['Date'] >= pd.to_datetime(date_range[0])) &
    (filtered_df['Date'] <= pd.to_datetime(date_range[1]))
]

# 3. KPIs (Top of the dashboard)
total_revenue = filtered_df['Revenue'].sum()
total_sales = filtered_df.shape[0] # Counts the number of rows

# Create three columns for the KPI cards
col1, col2, col3 = st.columns(3)

col1.metric("Total Revenue", f"${total_revenue:,.2f}")
col2.metric("Total Transactions", f"{total_sales}")
col3.metric("Top Product", filtered_df.groupby('Product')['Revenue'].sum().idxmax())

st.markdown("---")

# 4. CHARTS & VISUALIZATIONS
chart_col1, chart_col2 = st.columns(2)

with chart_col1:
    st.subheader("Revenue Trend Over Time")
    # Group by date to get daily totals for the line chart
    trend_df = filtered_df.groupby('Date', as_index=False)['Revenue'].sum()
    fig_line = px.line(trend_df, x='Date', y='Revenue', markers=True)
    st.plotly_chart(fig_line, use_container_width=True)

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with chart_col2:
    st.subheader("Revenue by Product Category")
    # Group by product for the bar chart
    bar_df = filtered_df.groupby('Product', as_index=False)
    ['Revenue'].sum().sort_values(by='Revenue', ascending=False)
    fig_bar = px.bar(bar_df, x='Product', y='Revenue', color='Product')
    st.plotly_chart(fig_bar, use_container_width=True)
```